

Pablo Pena

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EDUCATION

University of Texas at Austin

August 2024 – Fall 2027

B.S. in Mechanical Engineering Honors, Robotics Minor

GPA: 4.0

Relevant Coursework: Mechatronics, Robot Mechanism Design, Mechanics of Solids, Materials Engineering

SKILLS

CAD & Analysis: SolidWorks, Siemens NX, FEA, GD&T, Tolerance Stack-Up, NI Multisim

Manufacturing: Machine Shop (Drill Press, Bandsaw, Mill Maintenance), 3D Printing, Laser Cutting, SMT Soldering, Wire Harness Fabrication, CF/Kevlar Composite Layups

Programming: Python, MATLAB, C#, Arduino/Embedded, SQL, Java

PROFESSIONAL EXPERIENCE

BlackPearl Technology

June 2026 - August 2026

Mechanical Engineering Intern

The Woodlands, TX

- Designed the mechanical enclosure for a dual-PCB electromechanical DAQ device, working closely with electrical engineers to integrate BLE, LoRaWAN, and M2M hardware into a single ruggedized housing
- Engineered a torsional snap-fit mechanism enabling tool-free battery swaps, meeting drop protection (3 ft) and C1D1/Exd hazardous-area rating requirements for field deployment
- Took the design from CAD through prototype and into production, reducing component count by 67% through design consolidation and cutting assembly time by 60% through fixture and process improvements

NASA L'SPACE MCA

June 2025 - September 2025

Mechanical Payload Engineer

Remote

- Designed the mechanical architecture of a 250 kg Mars rover, developing a 6061 aluminum chassis and titanium rocker-bogie suspension for operation across extreme terrain and thermal environments
- Developed mechanical requirements and verification plans for 1,500 N static loads, 6,000 N shock loads, 2,000 Hz vibrations, and 45° inclines
- Engineered a precision 6-wheel rocker-bogie mobility system with independent wheel actuation and 4-wheel steering, enabling obstacle traversal up to 2× wheel diameter and continued operation after 2 wheel failures
- Developed CNC manufacturing and procurement plans for 4+ mechanical assemblies, estimating 10–15 business-day lead times with 35% schedule margin while evaluating in-house manufacturing for rapid iteration

PROJECTS

RC Plane — Texas Design Build Fly

August 2024 - May 2026

- Designed and hand-fabricated 4 fuselage structural prototypes using hybrid composite layups and vacuum bagging, iterating on CAD designs to optimize stiffness-to-weight ratio through repeated build-and-test cycles
- Designed carbon fiber/Kevlar landing gear with $\pm 45^\circ$ CF plies for torsional stiffness and 0° Kevlar plies for deflection resistance; validated structural layup quality through fit checks and assembly testing
- Flight-tested an 8G airframe structure carrying a 70 lb payload with no visible fuselage deformation

Battle Bot

January 2025 - May 2025

- Performed FEA-based stress analysis on the robot's frame and armor components under impact loading to inform material selection, then designed and fabricated 10 custom components using SolidWorks, 3D printing, and laser cutting
- Selected and integrated drive, actuation, and power subsystems, including DC motors, servo actuation, compression springs, and a 7.4 V LiPo battery pack, to improve durability and weapon performance

ACHIEVEMENTS & AWARDS

- Hispanic Scholarship Fund (HSF) Scholar

December 2026

- 6th Place Overall — AIAA Design/Build/Fly International Competition

April 2025